Task 1 - Data Cleaning and Preprocessing

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Dataset: Titanic (891 rows, 12 columns) **Tools:** Python, Pandas, Google Colab

What I did

First I loaded the dataset and used `df.info()`, `df.isnull().sum()` and `df.duplicated().sum()` to find the problems before changing anything. Then I fixed them one by one and ran the same checks again at the end to confirm everything worked.

Data quality issues and how I fixed them

#	Issue found	How I fixed it
1	Age had 177 missing values	Filled with the median age. I used median instead of mean because age has outliers.
2	Cabin had 687 missing values (about 77%)	Dropped the column, too much data was missing to fill it correctly.
3	Embarked had 2 missing values	Filled with the most common value (mode), since only 2 rows were affected.
4	Duplicate rows	Checked with duplicated(). Found 0 duplicates, so no rows were removed.
5	Text columns could have extra spaces and mixed casing	Used strip(), lower() and upper() to make Name, Sex and Embarked consistent.
6	Wrong data types (Age as float, Survived/Pclass as numbers)	Changed Age to int and Survived, Pclass, Sex, Embarked to category.
7	Fare had 116 outliers	Checked with the IQR method but kept them, because high fares are real first class tickets.

Result

- Null values after cleaning: 0
- Duplicate rows after cleaning: 0
- Columns: 12 before, 11 after (Cabin dropped)
- Files: titanic_cleaned.csv (cleaned data) and change_log.csv (every fix noted)

What I learned

Always check the data first before cleaning. Each column needs its own decision: some nulls can be filled, but some columns are better dropped. Outliers should not be removed automatically, because they can be real values.